Conditional probability and Bayes' theorem

Updating a belief when new evidence arrives — the most important topic in this step.

THE IDEA

P(A | B) means "the probability of A, *given* that B has happened". The bar means "given". New evidence changes the probability, and **Bayes' theorem** is the rule for updating it correctly.

> P(A | B) = P(B | A) · P(A) / P(B)

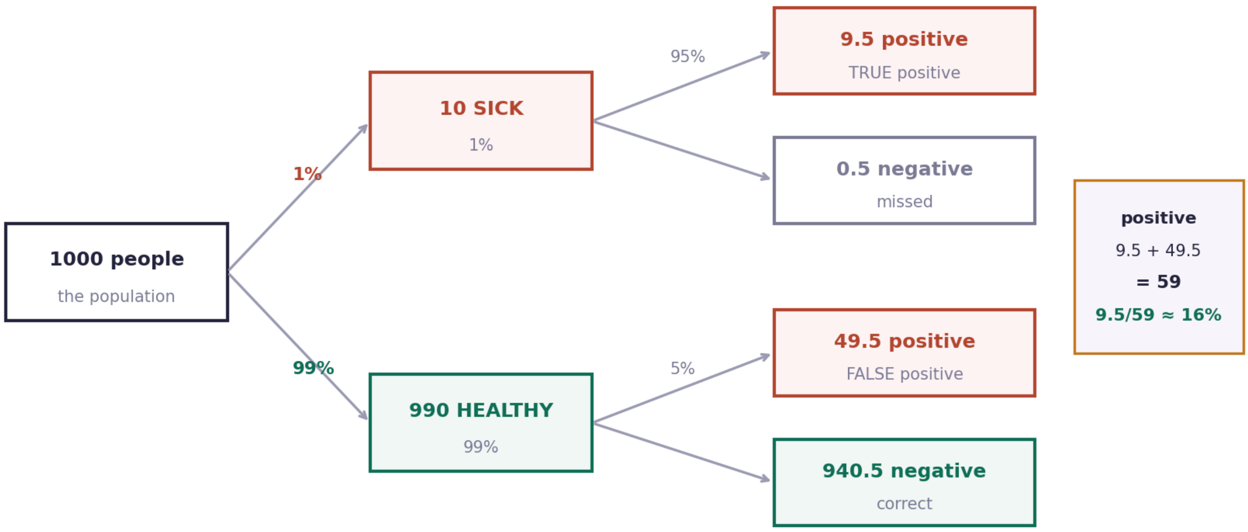
THE CLASSIC EXAMPLE

A disease affects 1% of people. A test is 95% accurate. You test positive. Most people answer "95%". The real answer is about **16%**.

THE EASIEST METHOD — IMAGINE 1000 PEOPLE

Rather than pushing symbols around, count actual people. It gives the same answer and is far harder to get wrong.

Why a 95% accurate test gives only a 16% chance of being sick



Group	Count	Test positive
Sick (1% of 1000)	10	95% → **9.5** true positives
Healthy (the other 99%)	990	5% → **49.5** false positives
Total positives	**9.5 + 49.5 = 59**	

So P(sick | positive) = 9.5 / 59 ≈ **0.16**.

WHY SO LOW

Because the healthy group is enormous. Five percent of 990 people outnumbers ninety-five percent of only 10 people. **The rarer the condition, the more the false positives dominate** — which is why a single positive result on a rare-disease screen is usually followed by a second, different test.

WORKED EXAMPLE

TOPIC 3 — CONDITIONAL PROBABILITY & BAYES

updating a belief when new evidence arrives

CONDITIONAL PROBABILITY

P(A | B) = the probability of A, GIVEN that B happened

the bar | means "given"

BAYES' THEOREM

P(A | B) = P(B | A) · P(A) / P(B)

THE CLASSIC EXAMPLE — a medical test

1% of people have a disease. the test is 95% accurate.

you test POSITIVE. do you have it ?

EASIEST METHOD — imagine 1000 people

SICK : 1% of 1000 = 10 people

95% test positive → 9.5 true positives

HEALTHY : the other 990 people

5% test positive anyway → 49.5 false positives

total positive tests = 9.5 + 49.5 = 59

P(sick | positive) = 9.5 / 59 ≈ 0.16

only about 16% — not 95%

WHY SO LOW ?

because the healthy group is huge. 5% of 990 people

outnumbers 95% of only 10 people.

the RARER the disease, the more false positives dominate.

ML LINK : this is why accuracy is a bad metric on rare events,
why precision and recall exist, and how spam filters and
Naive Bayes classifiers actually work.

> **Why it matters for ML:** this is why **accuracy is a misleading metric** on rare events — a model that always predicts "healthy" scores 99% on this population while catching nobody. It is the reason precision and recall exist, and it is the entire mechanism behind Naive Bayes classifiers and spam filters.

KEY POINTS FROM THIS TOPIC

Conditional	P(A | B) = probability of A given B
Bayes	P(A|B) = P(B|A) · P(A) / P(B)
Easiest method	imagine 1000 people and count
Key insight	rare condition → false positives dominate

TEST — Topic 3

1. 2% of a population has a condition. A test catches 90% of true cases. In 1000 people, how many are sick, and how many of those test positive? **→ 20 sick; 18 true positives**
2. Same setup, and the test wrongly flags 10% of healthy people. How many false positives among the 980 healthy? **→ 98**
3. Using both answers, what is P(sick | positive)? **→ 18 / (18 + 98) = 18/116 ≈ 0.155, about 16%**